

PhenoFusion3D

Final sprint review

Five plants recorded on 28 September 2026

3D reconstruction, manual trait comparison
and measured hyperspectral integration

Study scope and current evidence

Stage	Current result
Metric calibration	25 mm squares and 18 mm markers confirmed
D405 + L515 geometry	1,740,673 points before tissue cleanup
Five plant candidates	430,060 retained observed tissue points
Manual measurements	47 references across five plants
Hyperspectral samples	13,343 camera-space samples across FX10 + FX17

Independent physical accuracy and dense spectral correspondence remain research questions.

The known board size fixes the metric scale

25 mm

square edge

+2.8875%

D405 length correction

Board property	Confirmed value
Layout and dictionary	7 × 10 squares, DICT_4X4_50
Marker edge	18 mm
Independent L515 RGB motion factor	1.0296676

Scale correction preserves the observed surfaces. It cannot fill missing geometry.

D405 and L515 geometry fusion

- 1 Align the camera clouds
in a shared metric frame

1,719,001

D405 reference points

- 2 Check overlap against
observed source depths

+ 21,672

L515 additions

- 3 Keep source identities
with every retained point

1,740,673 observed points

Matched-soil median disagreement: 11.160 mm to 6.289 mm for the selected candidate.

Five cleaned plant candidates



430,060

retained tissue points across five plants

Manual heights and observed vertical spans

Plant	Manual height (cm)	Observed span (cm)	Span – manual (cm)
P1	53.5	48.16	-5.34
P2	51.7	51.33	-0.37
P3	52.8	50.49	-2.31
P4	52.3	44.81	-7.49
P5	31.8	24.99	-6.81

These spans do not certify stem-base-to-tip height. Missing endpoints can bias the comparison.

Thirteen conditional leaf comparisons

Dimension	Operator (cm)	3D chord (cm)	Difference (cm)
P3.O01.width	5.50	3.94	-1.56
P3.O02.width	7.60	9.27	+1.67
P5.O01.length	12.00	11.68	-0.32
P5.O03.length	11.90	9.80	-2.10

7 lengths and 6 widths across 8 organs in Plants 3 and 5

Natural-pose straight chords versus held-ruler references. These are exploratory comparisons.

The comparison retains its source evidence

P5.O01: same-organ provisional comparison
Natural-pose 3D chord versus held ruler; no geometry scaled to fit the ruler
Length: 3D 11.68 cm | operator 12.0 cm



P5.O01 length: 11.68 cm versus 12.0 cm

FX10 and FX17 measured spectra

Sensor	Wavelength range	Bands	Camera samples
FX10	397.66–1003.81 nm	224	7,982
FX17	935.61–1720.23 nm	224	5,361

FX10 supports exploratory NDVI and five other descriptors.

FX17 supports a recorded 1301.56 / 1449.68 nm signal ratio.

Unknown panel reflectance and dark reference limit physical interpretation.

Three locations with measured spectra

3 plant locations

2 sensor spectra
at each location

6 spectra in total



A wavelength change recolours the associated sample. Other surface points remain unassigned.

The earlier dense projection method



Earlier August prototype

10,485 projected spectral pixels
Local image registration plus depth

September extension

The tested height model fails
held-out leaf predictions.

Dense colour coverage alone does not establish correct spectral correspondence.

Dense projection diagnostic

REJECTED FOR MEASUREMENT



1,613 FX10 spectra

643 FX17 spectra

224 native bands
per sensor

**No accepted dense
assignment increment**

Withheld RGB errors of 7.04 and 8.24 px exceed the 3 px gate.

What the final sprint delivers

Available for review	Still requires evidence
Metric RGB-D geometry and source IDs	Complete surfaces and independent accuracy
Five cleaned plant candidates	Matched anatomical height and leaf validation
13 conditional leaf comparisons	Consistent anatomy and measurement protocol
FX10 and FX17 spectra and descriptors	Calibrated reflectance and dense 3D mapping
Offline software workflow and saved reports	New dataset and laboratory hardware testing

26 protected source files unchanged

101 targeted tests passed at the saved integration checkpoint

Next evidence needed for stronger results

Geometry coverage: visible stem base, fine tips and additional viewing angles

Trait references: labelled organs with matched endpoints and ruler geometry

Spectral geometry: common features at several heights across the scan

Radiometry: known white reflectance and confirmed dark capture

The next dataset should test the complete workflow without bespoke manual alignment.

Appendix: plane and leaf correspondence checks

Check	FX10	FX17
Held-out plane RMS	0.905–1.385 px	0.475–0.679 px
Leaf height model, point 1	10.08 px	7.24 px
Leaf height model, point 2	14.48 px	12.28 px
Leaf height model, point 3	174.70 px	123.43 px

The current height-dependent extension fails the held-out leaf test.

Appendix: demonstration order

- 1 September result overview and calibrated geometry
- 2 D405, L515, fused union and five cleaned plants
- 3 Manual measurements and explicit trait comparisons
- 4 FX10 and FX17 samples, wavelength and source quality
- 5 Sparse 3D associations and dense mapping evidence

Appendix: questions to expect

Does metric scale guarantee complete geometry?

Which manual traits use the same anatomical definition?

What evidence supports each spectral pixel-to-surface match?

How will the next recording test generalisation?

Which checks ran locally, and which need the lab?

Appendix: all P3 conditional comparisons

Dimension	Operator (cm)	3D chord (cm)	Difference (cm)
P3.O01.width	5.50	3.94	-1.56
P3.O02.length	9.80	9.12	-0.68
P3.O02.width	7.60	9.27	+1.67
P3.O03.length	7.50	7.69	+0.19
P3.O03.width	5.70	5.04	-0.66
P3.O04.length	11.00	11.50	+0.50
P3.O04.width	8.70	7.08	-1.62

Conditional same-organ comparison. Measurement protocol and endpoint conventions still matter.

Appendix: all P5 conditional comparisons

Dimension	Operator (cm)	3D chord (cm)	Difference (cm)
P5.O01.length	12.00	11.68	-0.32
P5.O01.width	9.90	9.76	-0.14
P5.O02.length	8.00	6.47	-1.53
P5.O03.length	11.90	9.80	-2.10
P5.O03.width	11.70	11.57	-0.13
P5.O05.length	13.20	11.98	-1.22

Conditional same-organ comparison. Measurement protocol and endpoint conventions still matter.

Appendix: signal normalisation and quality

Product	Definition	Interpretation
DN	Saved camera signal	Measured instrument value
Q	$(DN - D) / (W - D)$	Assumed offset correction
Q0	DN / W	Zero-offset sensitivity case
FX10 descriptors	NDVI, NDRE, GNDVI, PRI, PSRI, SIPI	Exploratory signal descriptors
FX17 descriptor	Q1301.56 / Q1449.68	Exploratory signal ratio

Invalid reference support remains missing. Colour alone cannot establish plant health.

Appendix: L515 additions and filtering

L515 review-region decision	Points
Input aligned review region	199,811
Duplicates suppressed	139,922
Conflicting or uncertain nonduplicates	38,217
Added to the fused scene	21,672
L515 points after plant cleanup	4,722

The fusion retains source observations. It does not infer hidden surfaces.

Appendix: retained points by plant

Plant	D405 points	L515 points	Total tissue points
P1	21,409	35	21,444
P2	72,518	268	72,786
P3	111,102	1,841	112,943
P4	68,043	196	68,239
P5	152,266	2,382	154,648

Point counts describe retained observations rather than biological completeness.

Appendix: offline operation and automation

Capability	Current scope
Offline software	Local processing and review with no AI service at runtime
Reusable input support	Validated files and explicit setup configurations
Reviewed inputs	Tissue regions, organ IDs, endpoints and spectral associations
Generalisation evidence	Further datasets and lab testing still required

Replayable processing is available. Fully automatic generalisation remains unestablished.

Appendix: local FX10 and FX17 tracking

Image-domain result	Value
Affine training inliers	54
One excluded green feature	0.589 native FX17 px
Tracks before final domain and depth checks	11,023
Unique rounded FX17 target pixels	6,984
Accepted new dense 3D assignments	0

Successful image tracking still needs a tested material-point link to the 3D geometry.